

# Can a carbon-aware VCG rule allocate shared GPU-hours better than first come, first served?

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**KEY MESSAGE** A published, carbon-aware VCG rule beats or ties all 720 FCFS arrival orders on value net of carbon (19.2 vs. 16.96 mean), cuts emissions per unit of value (0.72 vs. 0.81 kg), and keeps truthful reporting optimal, but serves fewer teams (4 vs. 4.93).

## 01 Question & model

### Research question & verified gap

**Question:** How should a university allocate scarce shared GPU-hours when each team privately knows its project value, but requests and emissions are public?

**Setting:** AI demand outpaces compute budgets; queues ignore both value and carbon. Stakeholders: research teams, cluster administrators, the public.

**Gap:** FAIR-Compute [5] studies fair, efficient allocation of federated research compute. We add an exactly solved VCG rule with carbon in the objective, plus a public audit.

### Strategic model & foundations

**Players:** six teams A–F; each reports  $r_i$  of its private value  $v_i \in [1, 10]$ .

**Public:** GPU request  $d_i$ , emissions  $e_i = 0.24 d_i$  kg CO<sub>2</sub>, capacity  $K = 100$ , carbon weight  $\lambda = 0.5$ .

**Timing:** one simultaneous sealed report (static, incomplete information).

**Payoff:**  $u_i = v_i - \lambda e_i - p_i$  if selected, else 0.

**Solution:** truthful  $r_i = v_i$  is dominant (DSIC, so also BNE) [1–4], because each team bears its own carbon penalty.

### Three complementary lenses

**Game theory:** Teams cannot benefit by lying about their value, so truthful reporting is the best strategy.

**Social choice:** We choose the set of teams that gives the highest total benefit while accounting for limited GPU capacity and carbon costs.

**Mechanism design:** Instead of first-come-first-served, we use a transparent scoring rule and VCG payments. Changing the carbon weight changes which teams receive GPUs.

### Auction or allocation comparison

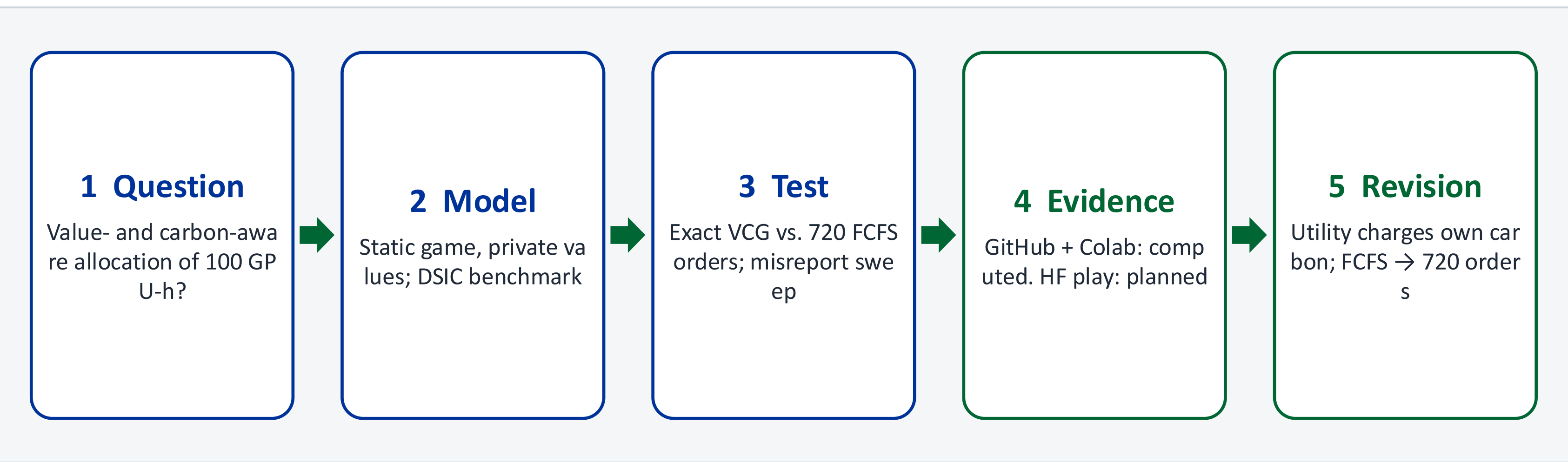
**Resource:** 100 GPU-hours; private values  $v_i$ ; sealed value reports  $r_i$ .

**Rules:** choose the feasible group  $S^*$  maximizing  $\sum (r_i - \lambda e_i)$  over all 64 groups; each winner pays  $p_i = W^*_{-i} - (\text{sum of the other winners' scores})$ , where  $W^*_{-i}$  is the best total without  $i$  ( $\times 10$  credits); one round.

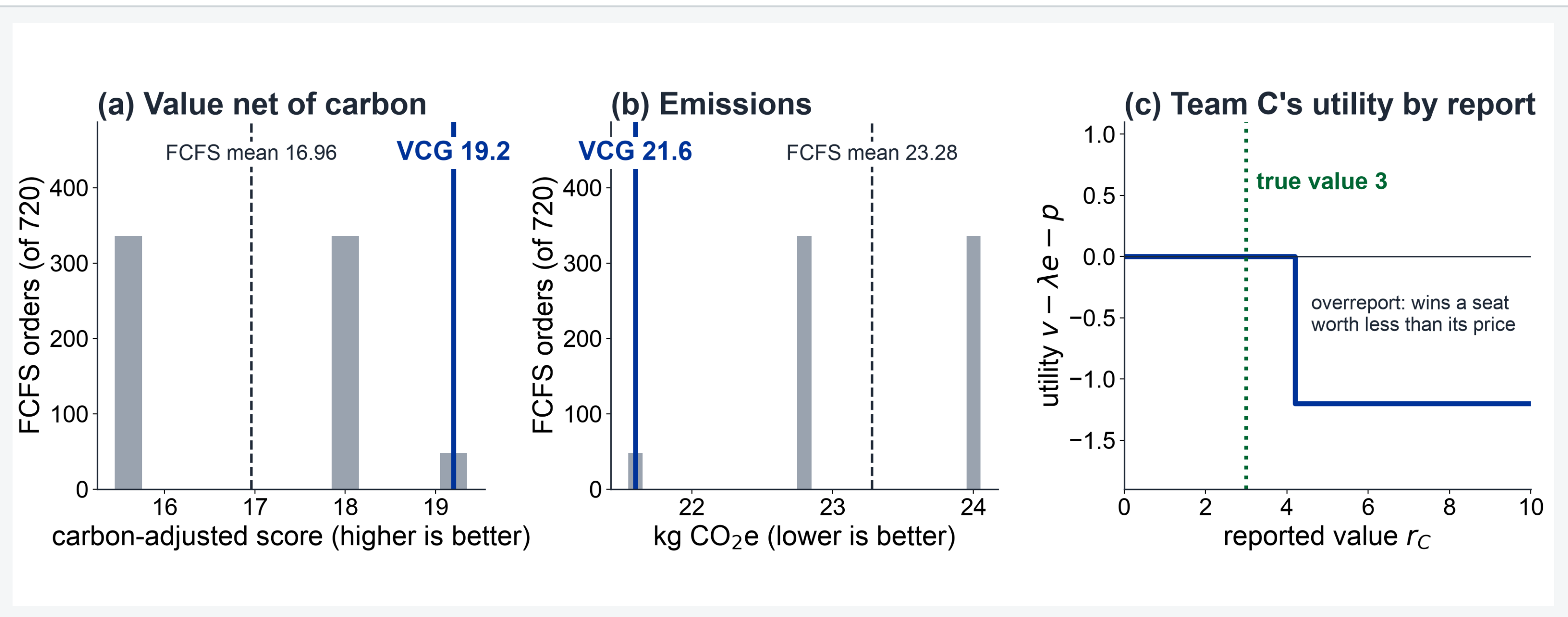
**Compared:** random-order FCFS (all 720 orders) vs. carbon-aware VCG.

**Analogy stops:** credits are non-transferable, values cannot be verified, and exact search does not scale.

## 02 Design & evidence



**Figure 1.** Research design. Tested (computed): exact VCG over all 64 groups vs. random-order FCFS over all 720 orders, and a misreport sweep for every team. Planned: classroom Hugging Face play recording reports, predictions, and reflections.



**Figure 2.** FCFS outcomes across all 720 arrival orders vs. VCG (a, b), and Team C's utility for every report (c).

**Interpretation:** VCG picks the highest-value, lowest-carbon group (A, B, D, E: value 30, 21.6 kg); no FCFS order beats it, but FCFS serves more teams on average.

**Verification:** python colab/run\_simulation.py reproduces the outputs; 5 unit tests pass; notebook 7/7 checks PASS; commit d93fc2e.

**Evidence status:** computed from one synthetic instance; behavioral play is planned, not observed.

## 03 Impact & dialogue

### Practical impact: a real-world case

**Case:** a university's shared GPU cluster (e.g., a lab pool at DKU); no adoption yet.

**Decision:** replace queue order with a published, re-runnable rule and a carbon weight  $\lambda$ .

**Benefits / risks:** high-value projects run first and allocations become auditable;  $\lambda$  can penalize compute-heavy research, and fewer teams are served.

**Before adoption:** credits need a real future cost; pilot reported values against ex-post outcomes.

**Correction:** public audit log;  $\lambda$  revised through reviews.

### Interdisciplinary evidence

#### Behavioral science

Players hold a private value, report, predict their selection, then see payments, utility, and a what-if curve; each copies a result line to a shared sheet.

**Boundary:** no plays collected yet; rivals are simulated.

#### Computer Science

Exact 64-group VCG and 720-order FCFS; python colab/run\_simulation.py reproduces the committed outputs; unit tests pass.

**Integration:** building the Space's what-if curve exposed that utility  $v_i - p_i$  rewards overreporting, so the model now charges each team its own carbon.

### Limitations & next test

**Limitation:** one synthetic instance;  $\lambda$  and 0.24 kg/GPU-h are assumptions; true values cannot be verified.

**Next test:** pilot comparing reported values with ex-post project outcomes at a scale beyond exact search.

**Question for reviewers:** how should  $\lambda$  be set, and who should set it?

SEP 28 Symposium, peer evaluation & case impact

SEP 30 Final update

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## Selected references

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- [5] Zachariadis, K. E., et al. (2026). FAIR-Compute: A roadmap for fair and efficient allocation of federated digital research infrastructure. arXiv:2607.28290

## OPEN MATERIALS

Hugging Face: [https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2\\_Team\\_A](https://huggingface.co/spaces/dku-comsci-econ206-2026/PS2_Team_A)  
GitHub: <https://github.com/Gihoone/COMPSCI206-PS2> (commit d93fc2e)

